



**MTH401- DIFFERENTIAL
EQUATIONS
(SOLVED MCQs)
FROM MIDTERM PAPERS
LECTURE (1-22)**



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CS619 & CS519

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Q1: Which of the following is the first order linear equation in unknown variable y?

A) $z \frac{dy}{dx} + (\sin x)y = \cos x$ (page48) (100% Sure)

D) $y \frac{dx}{dy} + (\sin y)x = \cos y$

A) $y \frac{dx}{dy} + (\sin y)x = \cos x$

A) $y \frac{dx}{dy} + (\sin x)x = \cos y$

Q2: Which of the following function would satisfy: $\frac{dy}{dx} = \frac{d^2y}{dx^2} = \dots$
 $= \frac{dny}{dx^n}$?

A) $y = Ae^x$ (Confirm) (100% Sure)

B) $y = Axe^x$

B) $y = Ae^{xx}$

B) $y = Ax^x e^{xx}$

Q3: If

$$y = 2 + x$$

, then which of the following is true for it?

A) Its annihilator is D

C) Its annihilator operator is D^2 (pag159) (100% Sure)

B) Its annihilator operator is D^3

B) Its annihilator operator is $D + 1$

Q4: For the non-exact differential equation $M(x,y)dx + N(x,y)dy = 0$,
if $\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} / N$ is a function of x, then the integrating factor is:

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A) function of x also (page40) (100% Sure)

D) function of y

D) constant

D) multi-variable function of both x and y

Q5: Which of the following are explicit solution of the differential equation: $\frac{dy}{dx} = -\frac{x}{y}$

A) $y = \pm \sqrt{4 + x}$

A) $y = \pm \sqrt{-4 + x^2}$

D) $y = \pm \sqrt{4 - x^2}$ (page16) (100% Sure)

D) $y = \pm \sqrt{-4 - x^2}$

Q6: For $f(x,y) = \frac{2x}{3y} + 7$, $f(t_x, t_y) = - - -$

A) $f(x,y)$

A) $t f(x,y)$ (page26) (100% Sure)

A) $t^2 f(x,y)$

D) $t^3 f(x,y)$

Q7: What is the annihilator operator of the function $g(x) = 4\sin x$?

A) $(D^2 - 1)$

D) $(D^2 + 4)$

D) $(D^2 - 4)$

D) $(D^2 + 1)$ (100% Sure)

Q8: The differential equation $\frac{dy}{dx} - y = y^3$

A) Bernoulli's (page55) (100% Sure)

A) Homogeneous

A) Cauchy

D) Bessel

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Q9: The integrating factor for the first order linear differential equation: $\frac{dy}{dx} + y \cot x = \sin^2 x$ is _____.

C) $\sin x$ (100% Sure)

D) $\cos x$

D) $e^{\sin x}$

D) $e^{\cos x}$

Q10: In exponential model for the population growth: $P(t) = P_0 e^{kt}$ If $k > 0$, then $\lim_{t \rightarrow \infty} P(t) =$

A) 0

B) 1

B) ∞ (page97) (100% Sure)

B) $-\infty$

Q11: The hook's law states that the force F is proportional to the _____.

A) Length

B) Elongation (page197) (100% Sure)

C) Weight

D) None of these

Q12: Separable form of the differential equation: $\frac{dy}{dx} = y - 1$ is _____, where $v = y - 1$.

A) $\frac{dx}{x} = dv$

B) $dx = v dv$

C) $\frac{dy}{v} = dx$ (page17) (100% Sure)

D) $\frac{dv}{v} = \frac{dx}{x}$

Q13: For $f(x, y) = \frac{2x}{3y} + 7$, $f(tx, ty) = \dots$

A) $f(x, y)$

B) $t f(x, y)$ (page26) (100% Sure)

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C) $t^2 f(x,y)$

D) $t^3 f(x,y)$

Q14: Which of the following is an equivalent form of the exact differential equation: $ydx + xdy = 0$?

A) $d \frac{x}{y} = 0$

B) $d \frac{y}{x} = 0$ (page33) (100% Sure)

C) $d(xy) = 0$

D) $d(x + y) = 0$

Q15: In the general solution of a separable differential equation is $\sin^{-1}y = \cos^{-1}x + c$, provided that $y(1/\sqrt{2}) = 1/\sqrt{2}$, then $c = \dots$.

A) $\pi/2$

B) $\pi/4$

C) $-\pi/2$

D) 0 (Confirm Solved) (100% Sure)

Q16: The differential equation of orthogonal trajectory to the family of curves $x - 2y = c$ is:

A) $dy/dx = -1/2$

B) $dy/dx = 1/2$

C) $dy/dx = 2$

D) $dy/dx = -2$ (Confirm) (100% Sure)

Q17: Classify the following differential equation $e^x dy/dx + 2y = 3xy$

A) Separable and not linear

B) Linear and not separable

C) Both separable and linear

D) Neither separable nor linear (Confirm Solved) (100% Sure)

Q18: The periodic time is given by:

A) $\omega/2\pi$

B) $2\pi/\omega$ (page201) (100% Sure)

C) $2\pi + \omega$

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D) $\frac{11}{w}$

Q19: If $x^2y^3dx + x^3y^2dy = 0$ has the equivalent form as $d(\frac{1}{3}x^3y^3) = 0$, then its solution is _____.

A) $x^3 + y^3 = c$

B) $x^3 - y^3 = c$

C) $x^3y^3 = c$

D) $\frac{x^3}{y^3} = c$

Q20: The family of parabolas $y^2 = 4ax$ are solutions of the differential equation: $\frac{dy}{dx} = \frac{2a}{y}$ for ----- value(s) of a.

A) infinite

B) finite

C) unique

D) no (Confirm) (100% Sure)

Q21: If $y = c_1e^{(-2+\sqrt{6})x} + c_2e^{(-2-\sqrt{6})x}$ is the complementary solution of:

A) $Ax + B$

D) $Ax^2 + Bx + C$

D) $Ax^3 + Bx^2 + Cx + D$ (Confirm) (100% Sure)

D) $(Ax^2 + Bx)2x^2$

Q22: If initial amount of radioactive isotope is 100g. What will be the amount at the end of 30 days such that $K=0.043$?

A) 371.415

B) 380.560

B) 363.297 (Confirm Solved) (100% Sure)

B) 360.351

Q23: If 2,3,5 are real roots of a differential equation, then the general solution is _____.

A) $y_c = c_1e^{2x} + c_2e^{3x} + c_3e^{5x}$ (confirm Solved) (100% Sure)

B) $y_c = c_1e^{2x} + c_2e^{-3x} + c_3e^{5x}$

B) $y_c = c_1e^{2x} + c_1e^{-3x} + c_1e^{5x}$

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B) $y_c = (C_1 + C_2 + C_3)e^{3x} + e^{2x}$

Q24: Wronskian $W(x, 2x) =$ _____.

- 0 (Confirm Solved) (100% Sure)
- 1
- -1
- None of these

Q25: Which of the following is the implicit solution of the differential equation: $\frac{dy}{dx} = -\frac{x}{y}$,

A) $x + y + 4 = 0$

B) $x^2 + y^2 - 4 = 0$ (page16) (100% Sure)

B) $x^2 - y^2 + 4 = 0$

B) $x^2 - y^2 - 4 = 0$

Q26: $ydx - y(\sin x)dy = 0$, is an example of _____ differential equation.

A) Exact (page33) (100% Sure)

B) Non-exact

B) Non-linear

B) Non-homogeneous

Q27: Which of the following is an example of ordinary differential equation?

A) $(y - x)dx + 4xdy = 0$ (page15) (100% Sure)

B) $\partial u / \partial y = -\partial u / \partial x$

B) $ax^2 + bx + c = 0, a \neq 0$

B) $\phi(x) = f(x) + \lambda \int k(x, t) \phi(t) dt$

Q28: A differential equation $M(x, y)dx + N(x, y)dy = 0$ is exact if there exists a multi-variable function $f(x, y)$ such that _____.

A) $df(x, y) = \partial f / \partial x dx + \partial f / \partial y dy$ (100% Sure)

B) $\int f(x, y)dx = \int \partial f / \partial x dx + \int \partial f / \partial y dy$

B) $f(x, y) = \partial f / \partial x dx + \partial f / \partial y dy$

B) $f(x, y) = \int \partial f / \partial x dx + \int \partial f / \partial y dy$

Q29: Which of the following is an initial value problem (IVP)?

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A) $d^2y/dx^2 + y = 0$

B) $d^2y/dx^2 + y = 0, x \in [-2, 2]$

B) $d^2y/dx^2 + y = 0, y(1) = -2$ and $y'(1) = -2$ (Confirm Solved) (100% Sure)

B) $d^2y/dx^2 + y = 0, y(1) = -2$ and $y'(\pi/2) = -2$

Q30: Which of the following would be a constant solution of the separable differential equation: $dy/dx = e^{x+y}$?

A) $y = 0$ (Confirm Solved) (100% Sure)

B) $y = 1$

B) $y = a \in \mathbb{R}$

B) No constant solution exist

Q31: The family of curve $x^2 + y^2 = C^2$ represents a family of circles centered at _____.

A) (1, 0)

D) (0, 1)

D) (0, 0) (page84) (100% Sure)

D) (1, 1)

Q32: The differential equation: $(x + 2y^3) dy/dx = y$ is linear in unknown:

A) variable y

B) multi- variable x and y

B) variable x (100% Sure)

B) dy/dx

Q33: For a system in simple harmonic motion which of the following is the time required to complete a cycle of motion?

A) Frequency

B) Amplitude

B) Period (page201) (100% Sure)

B) Revolution

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Q34: Suppose that $y_1, y_2, \dots, y_n$ are n solution of the homogeneous n th order differential equation. Then the solution is linearly dependent on an interval if and only if _____.

A) $W(y_1, y_2, \dots, y_n) = 0$ (page120) (100% Sure)

B) $W(y_1, y_2, \dots, y_n) = \infty$

C) $W(y_1, y_2, \dots, y_n) \neq 0$

D) $W(y_1, y_2, \dots, y_n) = -\infty$

Q35: If we substitute $u = y/x$ is differential equation.

A) $ue^u du = dx$

B) $ue^{(-u)} du = dx$ (page61) (100% Sure)

C) $ue^u du = dy$

D) $ue^{(-u)} du = dy$

Q36: Which of the following would be a particular solution of the differential equation: $dy/dx = 4$?

A) $y = 4x + a$

B) $y = ax + 4$

B) $y = ax + a$

B) $y = 4x + 4$ (Solved Confirm) (100% Sure)

Q37: A set of functions $\{f_1(x), f_2(x), f_3(x), \dots, f_n(x)\}$ is said to be _____ on an interval I if there exist constants.

A) Linear dependent (page111) (100% Sure)

B) Linear independent

Q38: Separable form $f(y)dy + g(x)dx = 0$, of the differential equation $x \sin y dx + (x^2 + 1) \cos y dy = 0$ is _____.

A) $\tan y dy + x/x^2 + 1 dx = 0$

B) $\cot y dy + x/x^2 + 1 dx = 0$ (Solved Confirm) (100% Sure)

B) $\tan y dy + x/x^2 - 1 dx = 0$

B) $\cot y dy + x/x^2 - 1 dx = 0$

Q39: The differential equation is $(1 + \ln xy)dx + (1 + x/y) dy = 0$ is exact because _____.

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A) $\partial M / \partial y = \partial N / \partial x = 1/x$

B) $\partial M / \partial x = \partial N / \partial y = 1/y$

B) $\partial M / \partial y = \partial N / \partial x = 1/y$ (Confirm Solved) (100% Sure)

B) $\partial M / \partial x = \partial N / \partial y = 1/x$

Q40: We can derive a differential equation governing the motion of a mass attached to spring when the Newton's second law combined with

.
A) Hook's law (page197) (100% Sure)

B) Newton's 3rd law